

VALJEVO, Serbia

WMO: 132690

Lat: 44.317N

Lon: 19.917E

Elev: 177

StdP: 99.22

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.4	-7.7	-12.9	1.3	-9.6	-10.3	1.6	-6.7	5.7	3.7	5.0	4.1	0.8	250	0.405

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.2	34.2	21.3	32.4	21.0	30.6	20.5	23.1	30.3	22.2	29.3	21.4	28.3	2.3	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.0	16.0	26.4	20.0	15.0	25.1	19.2	14.2	24.2	69.3	30.1	65.7	29.4	62.8	28.2	30.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 5.2	4.3	3.5	-14.9	36.7	3.9	2.2	-17.6	38.2	-19.9	39.5	-22.1	40.7	-24.9	42.3		
(5)			-15.1	24.6	3.9	1.7	-17.9	25.9	-20.1	26.9	-22.3	27.8	-25.1	29.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	12.3	1.3	3.2	7.4	12.4	16.9	20.6	22.5	22.5	17.7	12.5	7.5	2.4		
	DBStd	8.58	4.79	5.12	4.62	3.88	3.54	3.56	3.05	3.27	3.72	4.27	4.62	4.57		
	HDD10.0	959	271	194	107	21	1	0	0	0	0	24	102	239		
	HDD18.3	2629	528	424	339	179	69	18	4	5	57	185	327	494		
	CDD10.0	1797	1	3	26	94	216	319	388	388	230	102	26	3		
	CDD18.3	424	0	0	0	2	26	87	134	135	37	3	1	0		
	CDH23.3	4413	0	0	2	51	298	919	1369	1367	354	50	1	0		
	CDH26.7	1723	0	0	0	8	63	343	579	612	113	4	0	0		
(14) Wind	WSAvg	1.4	1.1	1.4	1.7	1.7	1.5	1.5	1.5	1.4	1.3	1.2	1.2	1.2		
(15) Precipitation	PrecAvg	795	49	47	59	61	85	100	76	64	70	65	58	60		
	PrecMax	1292	94	82	153	169	304	192	217	178	185	159	113	151		
	PrecMin	493	12	15	7	8	21	29	7	3	15	2	9	5		
	PrecStd	174	22	20	36	33	54	43	49	42	37	46	26	33		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.8	19.4	23.1	27.8	30.6	34.3	36.0	37.5	33.4	27.0	22.5	18.5		
		MCWB	10.2	10.9	12.7	16.3	18.9	21.7	21.1	21.5	20.1	17.3	14.7	12.2		
	2%	DB	13.0	16.3	20.6	24.5	28.4	32.3	33.9	34.5	30.0	24.8	19.9	14.3		
		MCWB	8.5	9.6	11.6	14.7	18.3	21.4	21.5	21.1	19.3	16.9	13.2	9.8		
	5%	DB	10.6	13.7	18.2	22.3	26.7	30.5	32.0	32.4	27.1	22.6	17.4	11.4		
		MCWB	7.3	8.5	10.9	13.7	17.8	20.8	21.2	20.7	18.2	15.8	12.2	8.2		
	10%	DB	8.1	11.0	15.8	20.2	24.8	28.6	30.0	30.2	24.9	20.2	15.0	8.8		
		MCWB	6.0	7.4	9.7	12.7	17.0	20.0	20.6	20.4	17.4	15.0	11.2	6.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.5	11.8	13.9	17.4	20.7	24.3	24.2	24.3	21.3	18.9	15.3	12.5		
		MCDB	14.9	17.4	21.2	25.4	27.4	31.0	31.8	31.1	29.7	25.0	20.8	17.5		
	2%	WB	9.1	10.1	12.4	15.6	19.4	22.7	22.9	23.0	20.2	17.6	13.8	10.0		
		MCDB	12.5	14.9	18.9	22.5	25.9	29.3	30.5	30.1	27.2	22.8	18.5	13.5		
	5%	WB	7.6	8.9	11.2	14.5	18.6	21.7	22.0	22.0	19.2	16.4	12.6	8.5		
		MCDB	10.2	13.0	17.2	20.5	24.9	28.2	29.2	29.2	25.1	21.2	16.5	11.1		
	10%	WB	6.2	7.6	10.1	13.5	17.6	20.8	21.2	21.2	18.2	15.4	11.4	6.9		
		MCDB	8.1	10.9	15.0	18.8	23.6	27.2	28.0	28.0	23.3	19.5	14.6	8.6		
(35) Mean Daily Temperature Range	MDBR		7.4	9.0	10.1	10.8	10.9	10.7	11.2	11.7	10.7	10.7	9.3	7.0		
	5% DB	MCDBR	12.6	14.9	15.5	15.6	14.8	14.6	15.5	16.5	15.3	14.9	13.7	12.1		
		MCWBR	8.4	8.5	7.7	7.1	6.2	5.5	5.2	5.4	6.0	7.4	7.7	8.0		
	5% WB	MCDBR	11.7	13.3	13.7	13.2	13.2	12.8	13.1	13.4	13.1	13.2	12.4	11.1		
		MCWBR	8.1	7.9	7.1	6.4	6.1	5.3	5.2	5.4	5.7	7.0	7.2	7.6		
(40) Clear-Sky Solar Irradiance	taub		0.337	0.366	0.401	0.459	0.450	0.445	0.446	0.454	0.432	0.393	0.352	0.334		
	taud		2.345	2.243	2.198	2.063	2.146	2.207	2.204	2.179	2.209	2.312	2.374	2.359		
	Ebn at Noon		767	805	825	805	825	830	823	801	784	767	743	730		
	Edh at Noon		81	106	126	157	149	141	139	138	124	98	78	73		
(44) All-Sky Solar Radiation	RadAvg		1.31	2.10	3.29	4.40	5.39	6.01	6.17	5.50	3.85	2.67	1.64	1.11		
	RadStd		0.19	0.27	0.39	0.49	0.52	0.58	0.48	0.54	0.42	0.31	0.15	0.17		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.76	N/A	N/A	N/A	+0.63	+0.65	N/A	N/A	N/A	N/A
(47) Regional (42 neighbors)		+0.66	N/A	N/A	N/A	+0.39	+0.45	N/A	-155	+109	+60

Nomenclature: See separate page